Projects Titles:

1) Serious Games for Wildfire Preparedness (Image name: wildfire.png)
2) Human-centered AI for Crew-autonomous Space Mission Planning (Image name: SpaceCHI.png)
3) Psychological Well-being and Human-centered AI (Image name: MH-AI.png)
4) Persona Driven Collaborative Storytelling (PDCS): Community Research (Image name: PDCS.png)
5) Reconceptualizing Human-AI experience from a Metaphysical Perspective (Image name: meta-physics.png)
6) Player experience and well-being in Massively Multiplayer Online Role-Playing Games (MMORPGs) (Image name: Sky children of the light.png)

1) Serious Games for Wildfire Preparedness

Research Overview:

In this project we developed a discourse-centered, triangulated methodology to understand how player to player communication in serious games fosters collaborative learning, shared understanding, and collective action for real-world wildfire challenges.

Research Objectives:

1) Analyze discourse patterns in collaborative serious games for wildfire preparedness
2) Identify and synthesize key communication strategies that facilitate knowledge transfer
3) Develop a methodology for evaluating collaborative learning in game-based disaster education
4) Create design guidelines for effective resilience based serious games

Contributions:

1) Fostering Collaborative Knowledge-Building and Resilience Through Player Discourse in Serious Games for Wildfire Preparedness **(paper accepted in JCSG'25)**. Dr. Magy Seif El-Nasr presented this research in December 2025 at the **Lucerne University of Applied Sciences and Arts** in **Rotkreuz, Switzerland.**
2) Talking Wildfire: How Discourse in Serious Games Shapes Collaborative Learning and Preparedness. **(FDG'26).** I presented this work at **Royal Danish Academy** in **Copenhagen, Denmark** in August 2026 (Virtually).

Here is the paper link for both the papers:

1) https://link.springer.com/chapter/10.1007/978-3-032-10518-9_11

2) https://dl.acm.org/doi/pdf/10.1145/3815598.3815644

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- Team: remove Darian lee from the team and also mention my position as a Research Scientist instead of research associate.
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2) Human-centered AI for Crew-autonomous Space Mission Planning

Research Overview:
In this project, we explored how artificial intelligence can support astronaut planning, scheduling, and execution during long-duration space missions where communication delays reduce real-time support from Earth. Through qualitative interviews with NASA subject matter experts involved in Playbook, we identified key challenges in mission planning, adaptation, and human-AI collaboration, and derived design implications for AI-assisted space mission systems.

Research Objectives:

1) Understand the challenges astronauts and planners face in mission planning and contingency replanning during communication-delayed missions

2) Examine how NASA subject matter experts and mission-planners perceive the role of AI in planning, scheduling, and execution workflows

3) Identify human-centered design implications for integrating AI without displacing human judgment in safety-critical contexts

4) Propose future research directions for AI-assisted systems that better support crew autonomy and resilience in long-duration exploration missions

Contributions:

1) Designing for Mars: Human-Centered AI, Resilience, and the Future of Crew-Autonomous Space Mission Planning **(SpaceCHI'26)**. Currently under print. I presented this paper at NASA Ames Research Center in September 2026.

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- Team: Shivam Shuka (University of California, Santa Cruz), Jimin Zheng (San Jose State University Research Foundation), Katherine Jiang (NASA Ames Research Center).
- External contributors: Shweta Sisodiya (Phd Student @UCSC), Shubham Gaur (Phd Student at UCSC)

3) Psychological Well-being and Human-centered AI

Research Overview:

In these projects, me and my research group are investigating how AI systems designed to provide emotional and psychological support shape human relationships and the fundamental need for relatedness. Drawing on psychological theories, Human-Computer Interaction (HCI), and Responsible AI, our research examines the risks of artificial intimacy/relatedness and relational displacement of human-connections in human-AI interactions. We propose a relationship-centered approach to AI design, alongside a computational framework to evaluate whether LLMs foster dependency on AI or encourage meaningful real-world human connections.

Research Objectives:

1. Investigate how emotionally supportive AI systems influence human relatedness, interpersonal relationships, and psychological well-being.
2. Conceptualize Relational Displacement in AI-mediated therapeutic relationships and develop relationship-centered design guidelines for responsible AI systems.
3. Develop **RELATE,** a computational evaluation framework to measure the relational orientation of LLMs across multi-turn emotionally supportive conversations.
4. Empirically evaluate how different LLMs and user personas influence inward-facing relational engagement versus outward-scaffolding toward real-world human connections.

5. Inform the design and evaluation of emotionally supportive AI systems that strengthen human agency, psychological well-being, and authentic interpersonal relationships.

Contributions:

1) S. Shukla, E. Chen, M. Roshanaei, M. Seif El-Nasr. Relationship-Centered Care: Relatedness and Responsible Design for Human Connections in Mental-Health Care. Position paper at the CHI 2026 Workshop: Toward Relationship-Centered Care with AI. [View Paper] Paper link: https://arxiv.org/pdf/2603.18375

2) S. Shukla, J. Kim, S. Gaur, M. Roshanaei, M. Seif El-Nasr. RELATE: An Evaluation Framework for measuring Relational Orientation of Large Language Models. [View Pre-print] Print link: https://arxiv.org/pdf/2610.09569

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- Team: Dr. Magy Seif El-Nasr (Advisor, UCSC), Dr. Emily Chen (Postdoctoral Researcher, UCSC), Dr. Mahnaz Roshanaei (Computational Social Scientist, Stanford University), Shivam Shukla (Phd Student, UCSC), Jessica Kim (NLP Researcher, UCSC), Shubham Gaur (Phd Student, UCSC)
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4) Persona Driven Collaborative Storytelling (PDCS): Community Research

Research Overview:

In this project, we developed **Persona-Driven Collaborative Storytelling (PDCS)**, a participatory research methodology that combines co-created personas, collaborative storytelling, and collective reflection to investigate sensitive technology use. Through a study with 25 Indian international students, we explored how LLMs support their academic, social, cultural, and emotional challenges while navigating life in the United States. Our research contributes a participatory approach to understanding marginalized communities' lived

experiences and generates insights into the role of AI in sociocultural adaptation and everyday support-seeking.

Research Objectives:

1. Develop a participatory research methodology that enables safe self-disclosure, collaborative storytelling, and participant-led interpretation of sensitive experiences.
2. Investigate how Indian international students use LLMs to navigate academic, cultural, social, and emotional challenges.
3. Examine the sociocultural factors influencing students' reliance on LLMs for everyday decision-making, adaptation, and support-seeking.
4. Evaluate participants' experiences with PDCS, particularly its ability to support memory recall, psychological safety, and meaningful participation.
5. Generate insights to inform the design of culturally responsive, inclusive, and human-centered AI systems for international student communities.

Contributions:

1) S. Sisodiya, S. Gotherwal, S. Shukla, E. Carstensdottir. Exploring Indian International Students' Experiences with LLMs. In Proceedings of the Human-Computer Interaction International Conference, (HCII'26), Springer Nature, 2026. [Under Print]

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- Team: Dr. Elin Carstensdottir (Advisor, UCSC), Shweta Sisodiya (Phd Student, UCSC), Shivam Shukla (Phd Student, UCSC)
- Remove support and collaboration

5) Reconceptualizing Human-AI experience from a Metaphysical Perspective

To be updated soon

6) Player experience and well-being in Massively Multiplayer Online Role-Playing Games (MMORPGs)

To be updated soon